

User Authentication, Role-Based Authorization, and Backend Security

Authentication

- JWT access token (short-lived) + refresh token (rotating recommended)
- Password hashing: bcrypt/argon2

Authorization (RBAC)

- Middleware checks:
 - `role == LEARNER` for learner features
 - `role == TEACHER` for teacher dashboard endpoints
 - ownership checks (`/profile/me`, `/progress/me`)

Security requirements

- HTTPS/TLS
- JWT authentication
- Secure password hashing